



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Quang Huy Nguyen
Mai 14th, 2026



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- We collected data from API
- We use data wrangling method to clean data
- We visualized data to see the relationships
- We see that all the launch sites are located near the coastline, highway, and railway.
- We found that the successful rate increases yearly.

Introduction

- Project background and context
 - Starlink is a satellite internet constellation providing satellite Internet access. Sending manned missions to Space. One reason SpaceX can do this is the rocket launches are relatively inexpensive. SpaceX advertises Falcon 9 rocket launches on its website with a cost of 62 million dollars; other providers cost upwards of 165 million dollars each, much of the savings is because SpaceX can reuse the first stage. Therefore, if we can determine if the first stage will land, we can determine the cost of a launch. SpaceX's Falcon 9 launch like regular rockets.
 - The first stage is quite large and expensive. Unlike other rocket providers, SpaceX's Falcon 9 Can recover the first stage. Sometimes the first stage does not land. Sometimes it will crash as shown in this clip. Other times, Space X will sacrifice the first stage due to the mission parameters like payload, orbit, and customer.
- Problems you want to find answers
 - To determine the price of each launch
 - To gathering information about SpaceX and to create dashboard for the team
 - To determine if SpaceX will reuse the first stage by training a machine learning model.

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - We gathered SpaceX launch data from an API, particularly SpaceX REST API
- Perform data wrangling
 - Calculated the number of launches per site
 - Calculate the number and occurrence of each orbit
 - Calculate the number and occurrence of mission outcome of the orbits
 - Create a landing outcome label from Outcome column

Methodology

Executive Summary

- Perform exploratory data analysis (EDA) using visualization and SQL
 - Perform exploratory Data Analysis and Feature Engineering using Pandas and Matplotlib
- Perform interactive visual analytics using Folium and Plotly Dash
 - Mark all launch site on a map
 - Mark the success/failed launches for each site on the map
 - Calculate the distances between a launch site to its proximities
 - Create dashboard to show the figures:
 - Total successful launches count for all sites
 - Success and Failed counts for the selected site
 - Scatter chart to show the correlation between payload and launch success

Methodology

Executive Summary

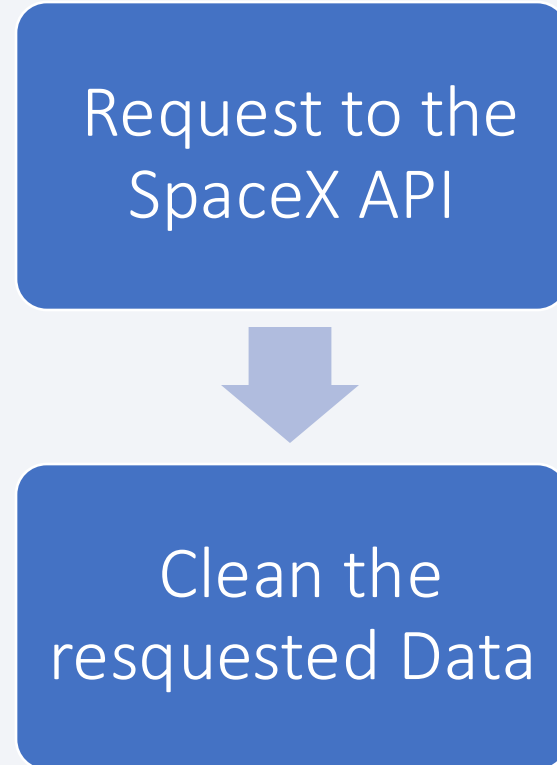
- Perform predictive analysis using classification models
 - Create a column for the class
 - Standardized the data
 - Spilt into training and test data
 - Hyperparameters for SVM, Classification Trees, and Logistic Regression using GridSearchCV
 - Find the methods performs best using test data

Data Collection

- Describe how data sets were collected.
- You need to present your data collection process use key phrases and flowcharts

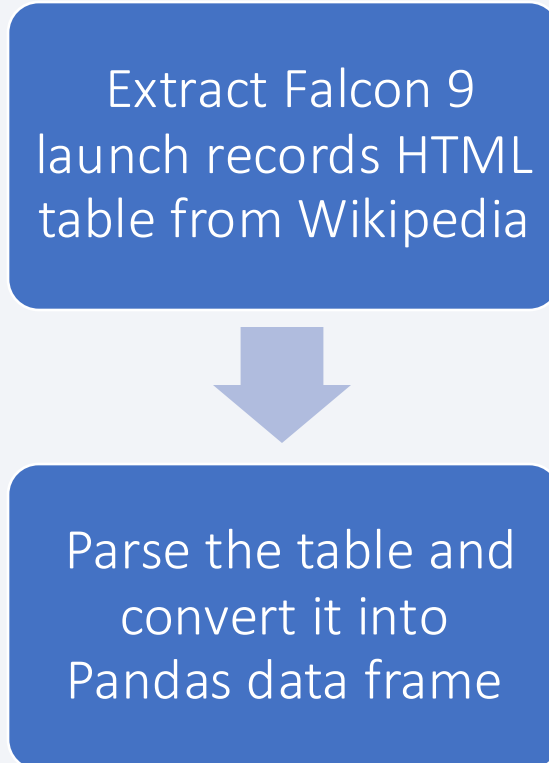
Data Collection – SpaceX API

- Present your data collection with SpaceX REST calls using key phrases and flowcharts
- Github link:
- https://github.com/nguyenquanghuyvn/Applied_Data_Science_Capstone/blob/main/Notebooks/1_1_done_practice_jupyter-labs-spacex-data-collection-api-v2.ipynb



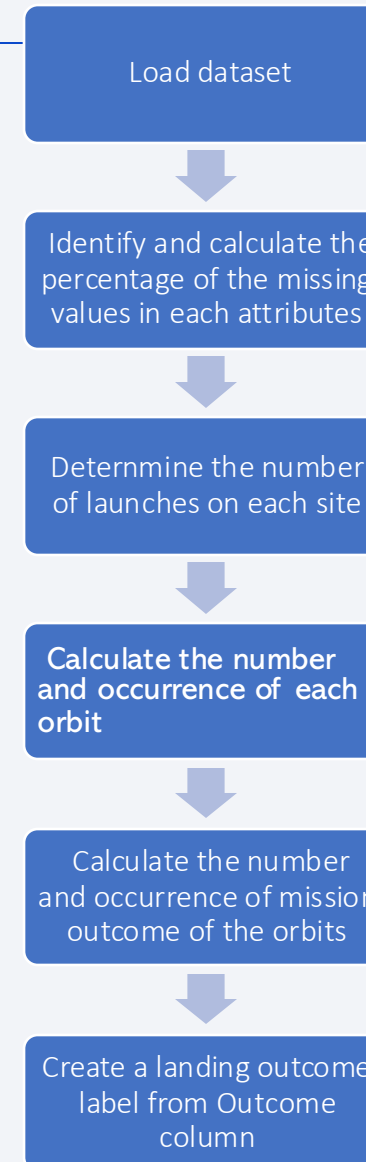
Data Collection - Scraping

- Present your web scraping process using key phrases and flowcharts
- Github link:
- https://github.com/nguyenquanghuyvn/Applied_Data_Science_Capstone/blob/main/Notebooks/1_2_practice_jupyter-labs-webscraping.ipynb



Data Wrangling

- Calculate the number of launches on each site
- Calculate the number and occurrence of each orbit
- Calculate the number and occurrence of mission outcome per orbit type
- Create a landing outcome label from outcome column:
 - 1 means the booster successfully landed
 - 0 means it was unsuccessfully landed
- Github link:
 - https://github.com/nguyenquanghuyvn/Applied-Data-Science-Capstone/blob/main/Notebooks/1_3_Practice_data_wrangling.ipynb



EDA with Data Visualization

- We visualized the relationship between Flight Number and Launch Site to see more details of the launch record.
- We visualized the relationship between Payload Mass and Launch Site to see whether there are any relationship between launch sites and there payload mass.
- We visualized the relationship between success rate of each orbit type by using bar chart to identify which orbits have the highest success rates.
- We visualized the relationship between Flight Number and Orbit type to observe if the success rate relates to the number of flights and orbit type.
- We visualized the relationship between Payload Mass and Orbit type
- We visualized the launch success yearly trend
- Github

link: https://github.com/nguyenquanghuyvn/Applied_Data_Science_Capstone/blob/main/Notebooks/2_2_Done_Practice_eda-dataviz-v2.ipynb

EDA with SQL

- Using bullet point format, summarize the SQL queries you performed
- %sql select distinct "Launch_Site" from SPACEXTABLE
- %sql select * from SPACEXTABLE where "Launch_Site" like "%CCA%" Limit 5;
- %%sql select sum("PAYLOAD_MASS__KG_") as TOAL_PAYLOAD_MASS from SPACEXTABLE WHERE Customer like "%NASA%"
- %%sql select AVG("PAYLOAD_MASS__KG_") as AVG_PAYLOAD_MASS from SPACEXTABLE where "Booster_Version" like "%F9 V1.1%"
- %%sql select distinct "Landing_Outcome" from SPACEXTABLE
- %sql select min(Date) from SPACEXTABLE where "Landing_Outcome" = "Success (ground pad)"

EDA with SQL

- Using bullet point format, summarize the SQL queries you performed
- `%%sql SELECT "Booster_Version", "Landing_Outcome", "PAYLOAD_MASS__KG_"
FROM SPACEXTABLE WHERE ("Landing_Outcome"="Success (drone ship)") AND
("PAYLOAD_MASS__KG_" BETWEEN 4000 AND 6000)`
- `%%sql SELECT Mission_Outcome, COUNT(*) AS TOTAL_NUMBER FROM SPACEXTABLE GROUP BY
Mission_Outcome`
- `%sql SELECT "Booster_Version" FROM SPACEXTABLE WHERE "PAYLOAD_MASS__KG_" = (SELECT
MAX("PAYLOAD_MASS__KG_") FROM SPACEXTABLE)`
- `%%sql SELECT substr(DATE, 6, 2) AS MONTH, "Landing_Outcome", "Booster_Version", "Launch_site"
FROM SPACEXTABLE WHERE "Landing_Outcome" = "Failure (drone ship)" AND substr(DATE, 1, 4) =
"2015";`
- `%%sql SELECT "Landing_Outcome", DATE, COUNT(*) AS TOTAL FROM SPACEXTABLE WHERE DATE
BETWEEN "2010-06-04" AND "2017-03-20" GROUP BY "Landing_Outcome" ORDER BY TOTAL DESC`

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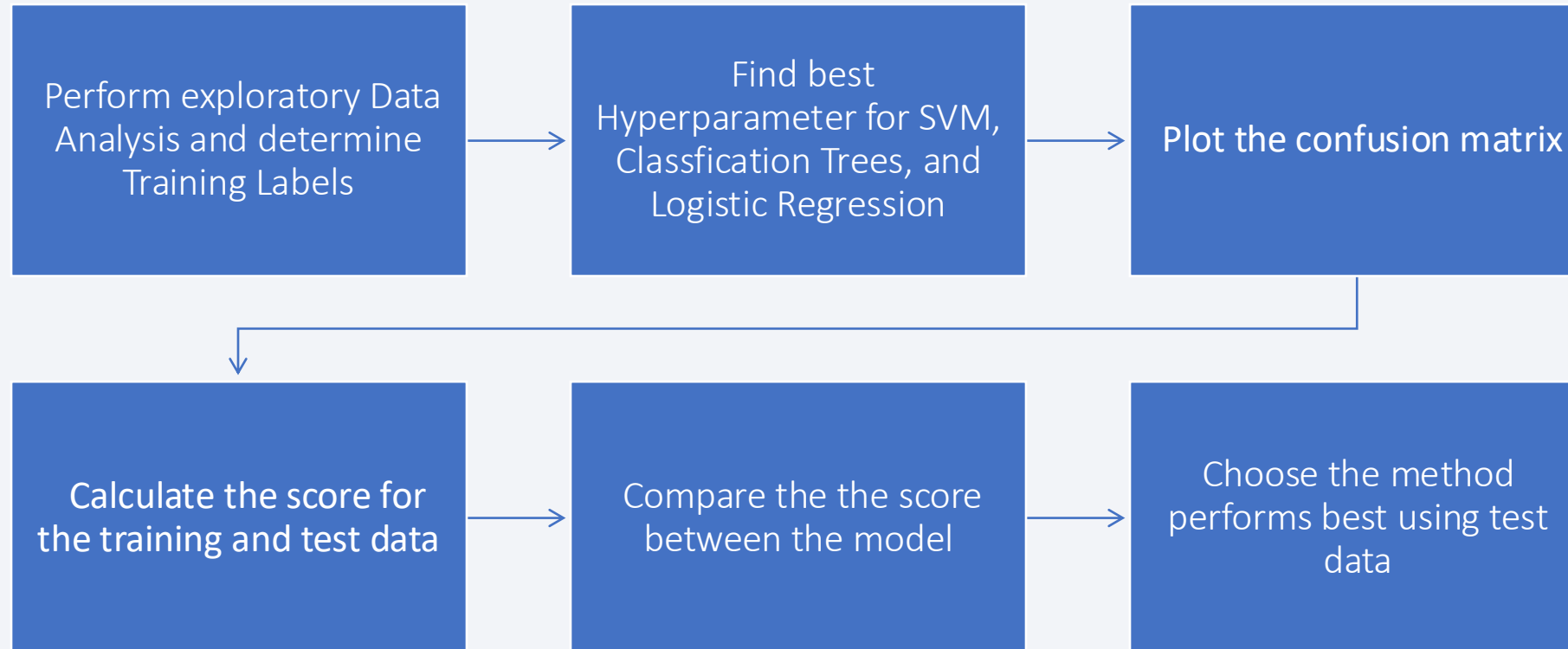
Build an Interactive Map with Folium

- The map objects are:
 - Circles to identify Nasa Johnson Space Centers and four launch sites.
 - Lines to measure the distances from launch sites to coastlines, railroads, and highways.
- Add the GitHub
URL: https://github.com/nguyenquanghuyvn/Applied_Data_Science_Capstone/blob/main/Notebooks/3_1_Practice-launch-site-location-v2.ipynb

Build a Dashboard with Plotly Dash

- Summarize what plots/graphs and interactions you have added to a dashboard
- In the dashboard, we added the pie charts to show
 - The total success launches by site
 - For the selected site, the success rate and failure rate
- The scatter plot for the relationship between Payload and success for all sites.
- We add the interactions
 - Launch-Site drop-down to select the All sites or specific site
 - Callback function to render success pie chart on the selected site dropdown
 - Ranger slider to Select Payload
 - Callback to render the success payload scatter plot
- Add the GitHub
URL: https://github.com/nguyenquanghuyvn/Applied_Data_Science_Capstone/blob/main/Notebooks/3_2_practice_spacex_dash_app.py

Predictive Analysis (Classification)

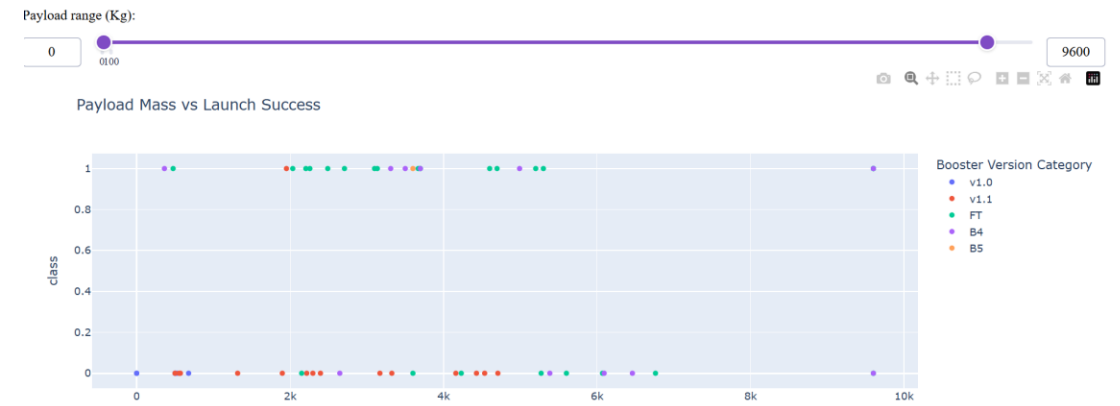
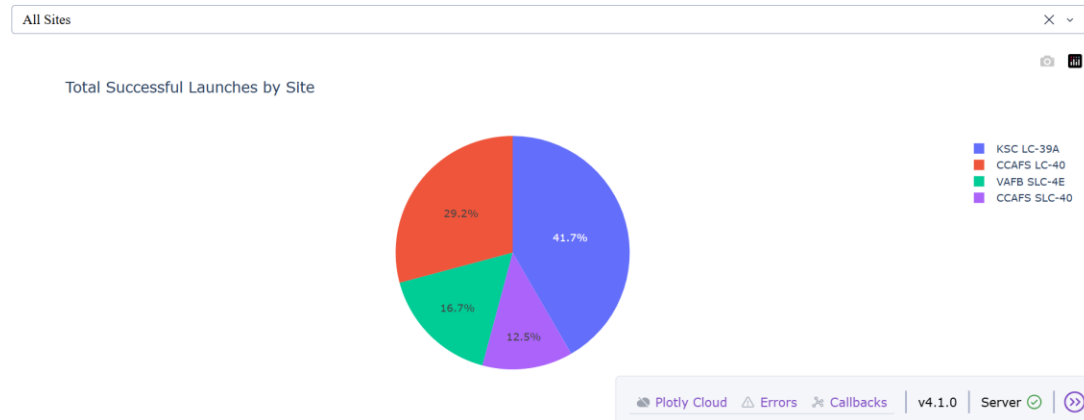


- Github
link: https://github.com/nguyenquanghuyvn/Applied_Data_Science_Capstone/blob/main/Notebooks/4_1_DONE_SpaceX-Machine-Learning-Prediction-Part-5-v1.ipynb

Results

- Exploratory data analysis results
 - We see that as the flight number increases, the first stage is more likely to land successfully. The payload mass also appears to be a factor; even with more massive payloads, the first stage often returns successfully.
 - We found for the VAFB-SLC launchsite there are no rockets launched for heavypayload mass(greater than 10000)
 - ES-L, GEO, GEO, and SSO are the orbits with the highest success rates.
 - In the LEO orbit, success seems to be related to the number of flights. Conversely, in the GTO orbit, there appears to be no relationship between flight number and success.
 - With heavy payloads the successful landing or positive landing rate are more for Polar,LEO and ISS.
 - The sucess rate since 2013 kept increasing till 2020
- Interactive analytics demo in screenshots
- Predictive analysis results

SpaceX Launch Records Dashboard

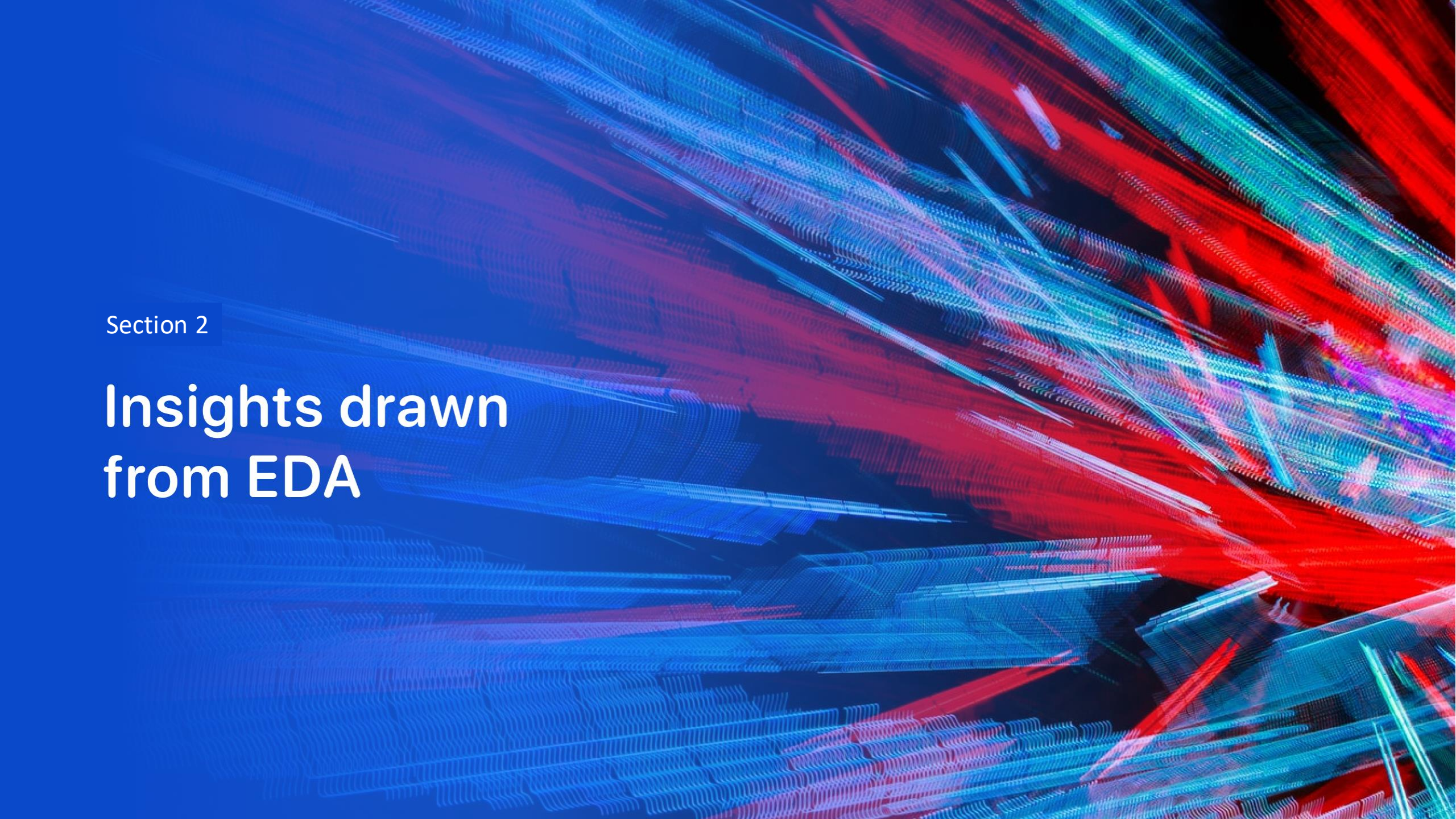


Results

- Interactive analytics demo in screenshots

Results

- Predictive analysis results
 - The methods predicted 12 true positives, 3 false positives, and three true negatives.
 - All the methods have the test scores are about 83.33%

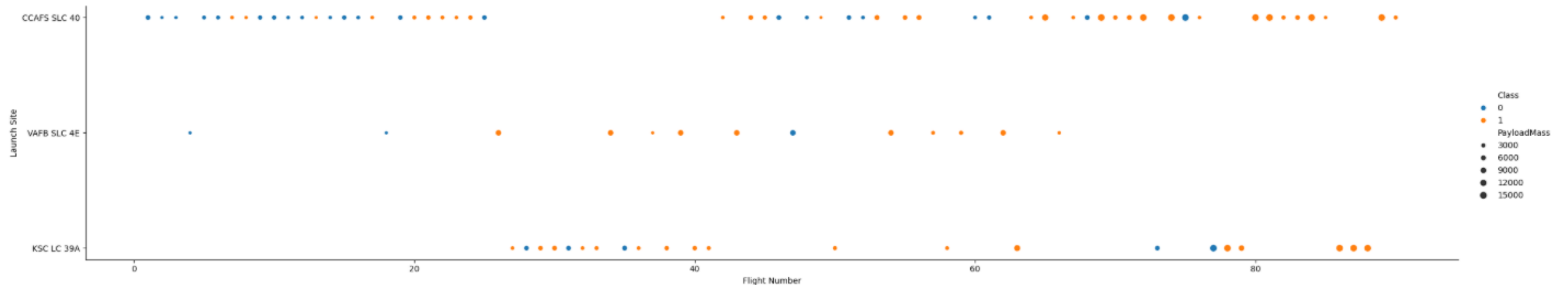
The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of blue and red, creating a sense of motion or data flow. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is high-tech and digital.

Section 2

Insights drawn from EDA

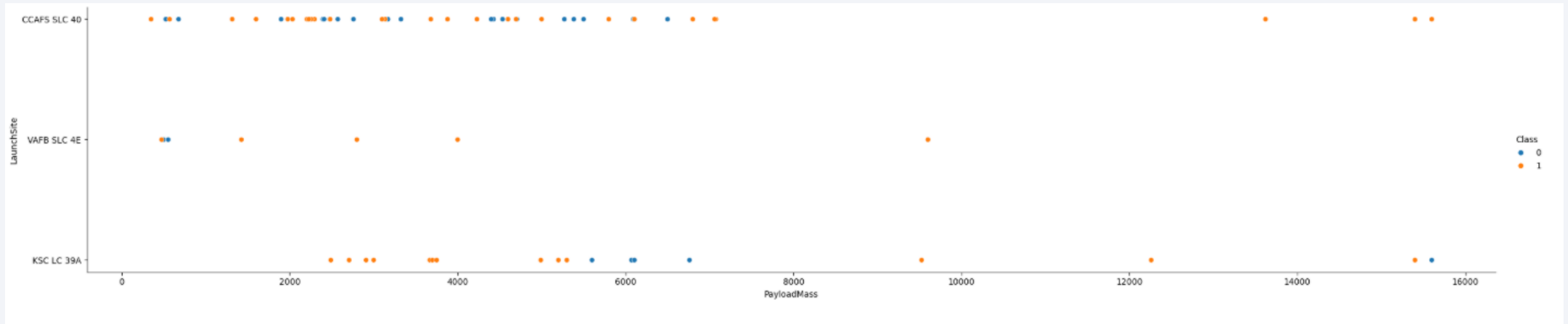
Flight Number vs. Launch Site

The most frequent launch site is CCAFS SLC 40



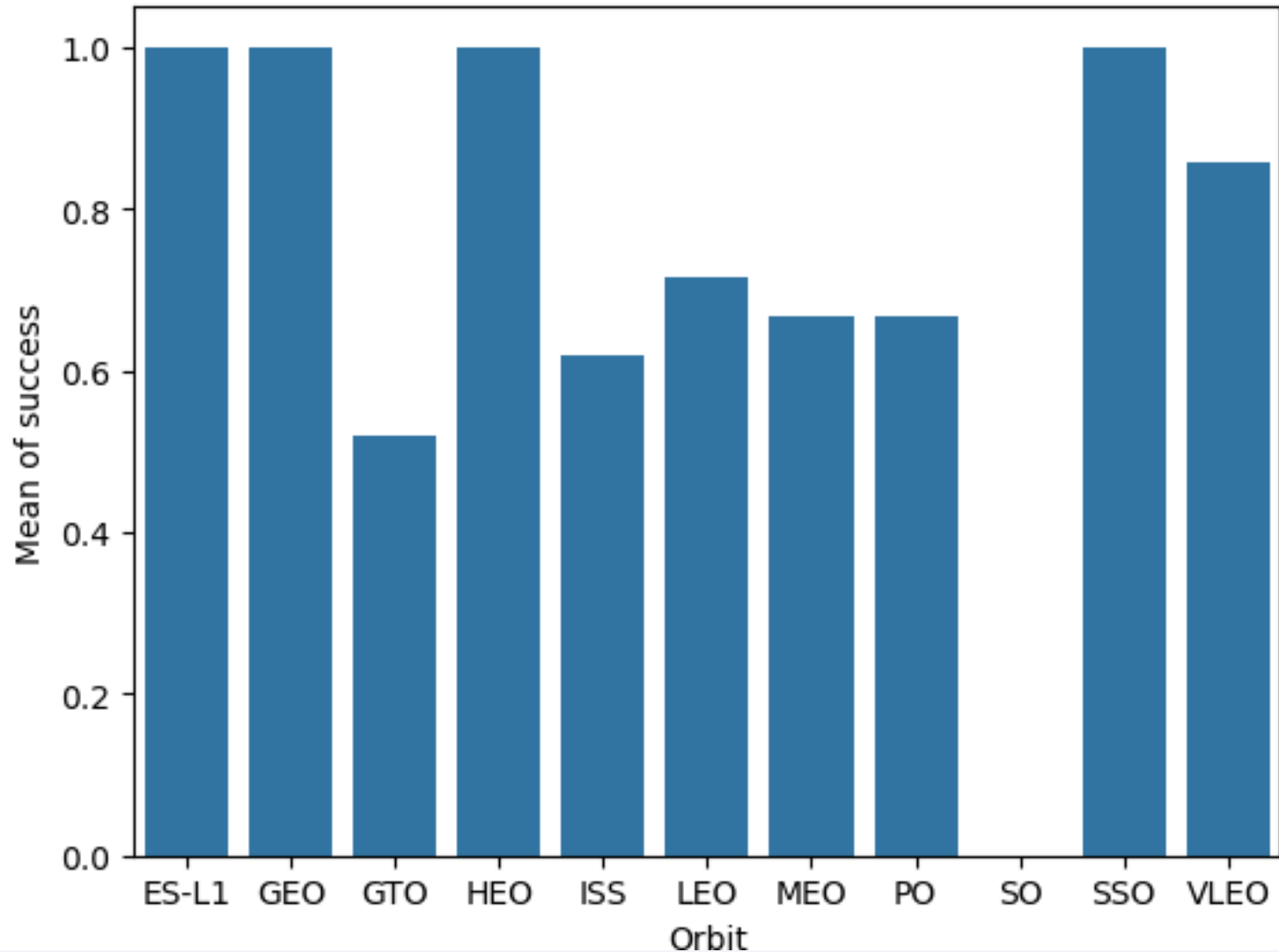
Payload vs. Launch Site

- At the VAFB-SLC launch site, there have been no rockets launched with a heavy payload mass (greater than 10,000).



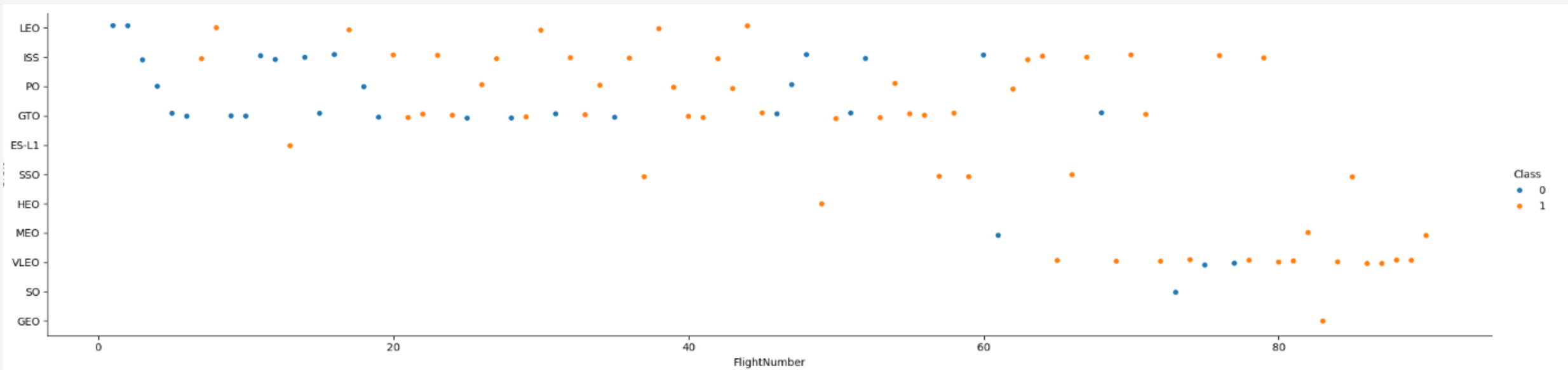
Success Rate vs. Orbit Type

The orbit ES-L, GEO, HEO, SSO have the highest successful rates.



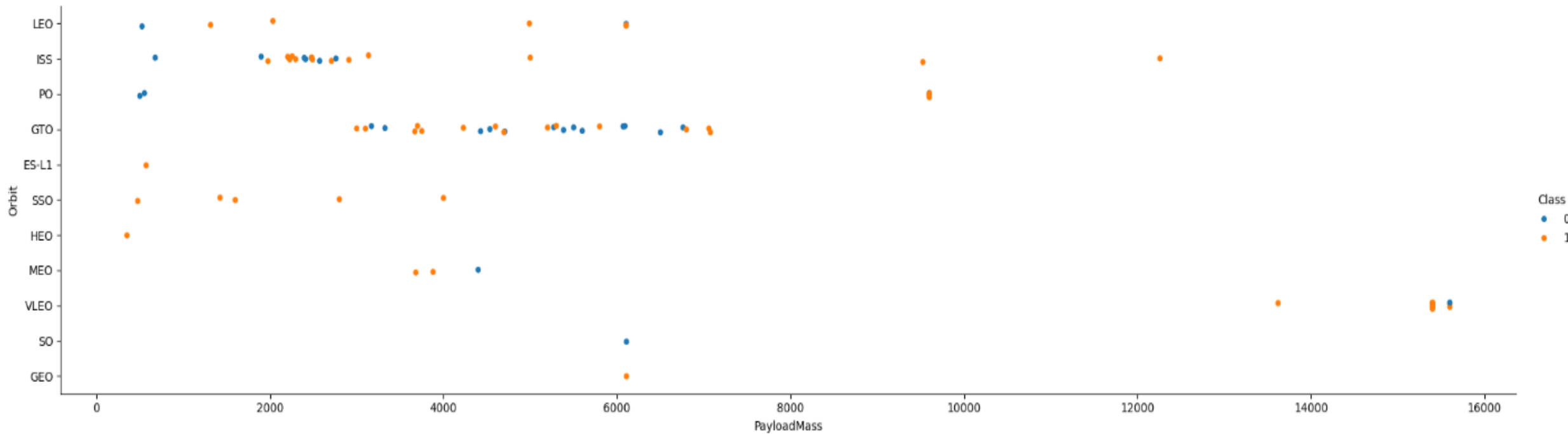
Flight Number vs. Orbit Type

- In the LEO orbit, success seems to be related to the number of flights. Conversely, in the GTO orbit, there appears to be no relationship between flight number and success.



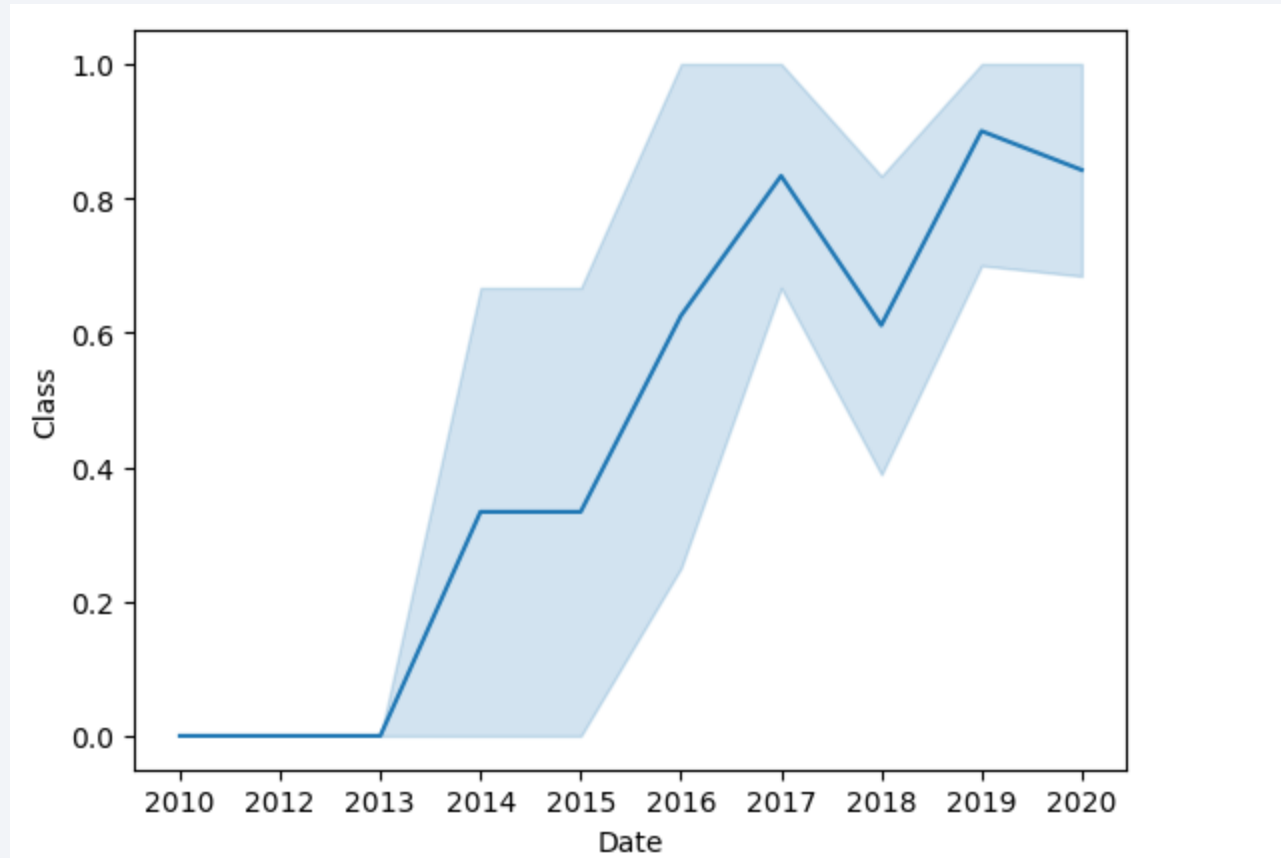
Payload vs. Orbit Type

- With heavy payloads the successful landing or positive landing rate are more for Polar, LEO and ISS.



Launch Success Yearly Trend

- The success rate since 2013 kept increasing till 2020.



All Launch Site Names

- The query used is %sql select distinct "Launch_Site" from SPACEXTABLE
- The names of the unique launch sites are CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, and CCAFS SLC-40.

Launch Site Names Begin with 'CCA'

- Find 5 records where launch sites begin with `CCA`:

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Query: %sql select * from SPACEXTABLE where "Launch_Site" like "%CCA%" Limit 5;

Total Payload Mass

- Calculate the total payload carried by boosters from NASA
- `%%sql select sum("PAYLOAD_MASS__KG_") as TOAL_PAYLOAD_MASS from SPACEXTABLE WHERE Customer like "%NASA%"`

TOAL_PAYLOAD_MASS

107010

Average Payload Mass by F9 v1.1

- Calculate the average payload mass carried by booster version F9 v1.1
- `%%sql select AVG("PAYLOAD_MASS__KG_") as AVG_PAYLOAD_MASS from SPACEXTABLE where "Booster_Version" like "%F9 V1.1%"`

AVG_PAYLOAD_MASS

2534.6666666666665

First Successful Ground Landing Date

Landing_Outcome

Failure (parachute)

No attempt

Uncontrolled (ocean)

Controlled (ocean)

Failure (drone ship)

Precluded (drone ship)

Success (ground pad)

Success (drone ship)

Success

Failure

No attempt

- Find the dates of the first successful landing outcome on ground pad
- `%%sql select distinct "Landing_Outcome" from SPACEXTABLE`

Successful Drone Ship Landing with Payload between 4000 and 6000

- List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000
- `%%sql SELECT "Booster_Version", "Landing_Outcome", "PAYLOAD_MASS_KG_" FROM SPACEXTABLE WHERE ("Landing_Outcome"="Success (drone ship)") AND ("PAYLOAD_MASS_KG_" BETWEEN 4000 AND 6000)`

Booster_Version	Landing_Outcome	PAYLOAD_MASS_KG_
F9 FT B1022	Success (drone ship)	4696
F9 FT B1026	Success (drone ship)	4600
F9 FT B1021.2	Success (drone ship)	5300
F9 FT B1031.2	Success (drone ship)	5200

Total Number of Successful and Failure Mission Outcomes

- Calculate the total number of successful and failure mission outcomes
- `%%sql SELECT Mission_Outcome, COUNT(*) AS TOTAL_NUMBER FROM SPACEXTABLE GROUP BY Mission_Outcome`

Mission_Outcome	TOTAL_NUMBER
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

Booster_Version

F9 B5 B1048.4

F9 B5 B1049.4

F9 B5 B1051.3

F9 B5 B1056.4

F9 B5 B1048.5

F9 B5 B1051.4

F9 B5 B1049.5

F9 B5 B1060.2

F9 B5 B1058.3

F9 B5 B1051.6

F9 B5 B1060.3

F9 B5 B1049.7

- List the names of the booster which have carried the maximum payload mass
- %sql SELECT "Booster_Version" FROM SPACEXTABLE WHERE "PAYLOAD_MASS__KG_" = (SELECT MAX("PAYLOAD_MASS__KG_") FROM SPACEXTABLE)

2015 Launch Records

- List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015
- %%sql
- SELECT substr(DATE, 6, 2) AS MONTH, "Landing_Outcome", "Booster_Version", "Launch_site"
- FROM SPACEXTABLE
- WHERE "Landing_Outcome" = "Failure (drone ship)"
- AND YEAR(DATE) = 2015

MONTH	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

- `%%sql`
- `SELECT "Landing_Outcome",
DATE, COUNT(*) AS TOTAL`
- `FROM SPACEXTABLE`
- `WHERE DATE BETWEEN "2010-06-04" AND "2017-03-20"`
- `GROUP BY "Landing_Outcome"`

Landing_Outcome	Date	TOTAL
No attempt	2012-05-22	10
Success (drone ship)	2016-04-08	5
Failure (drone ship)	2015-01-10	5
Success (ground pad)	2015-12-22	3
Controlled (ocean)	2014-04-18	3
Uncontrolled (ocean)	2013-09-29	2
Failure (parachute)	2010-06-04	2
Precluded (drone ship)	2015-06-28	1

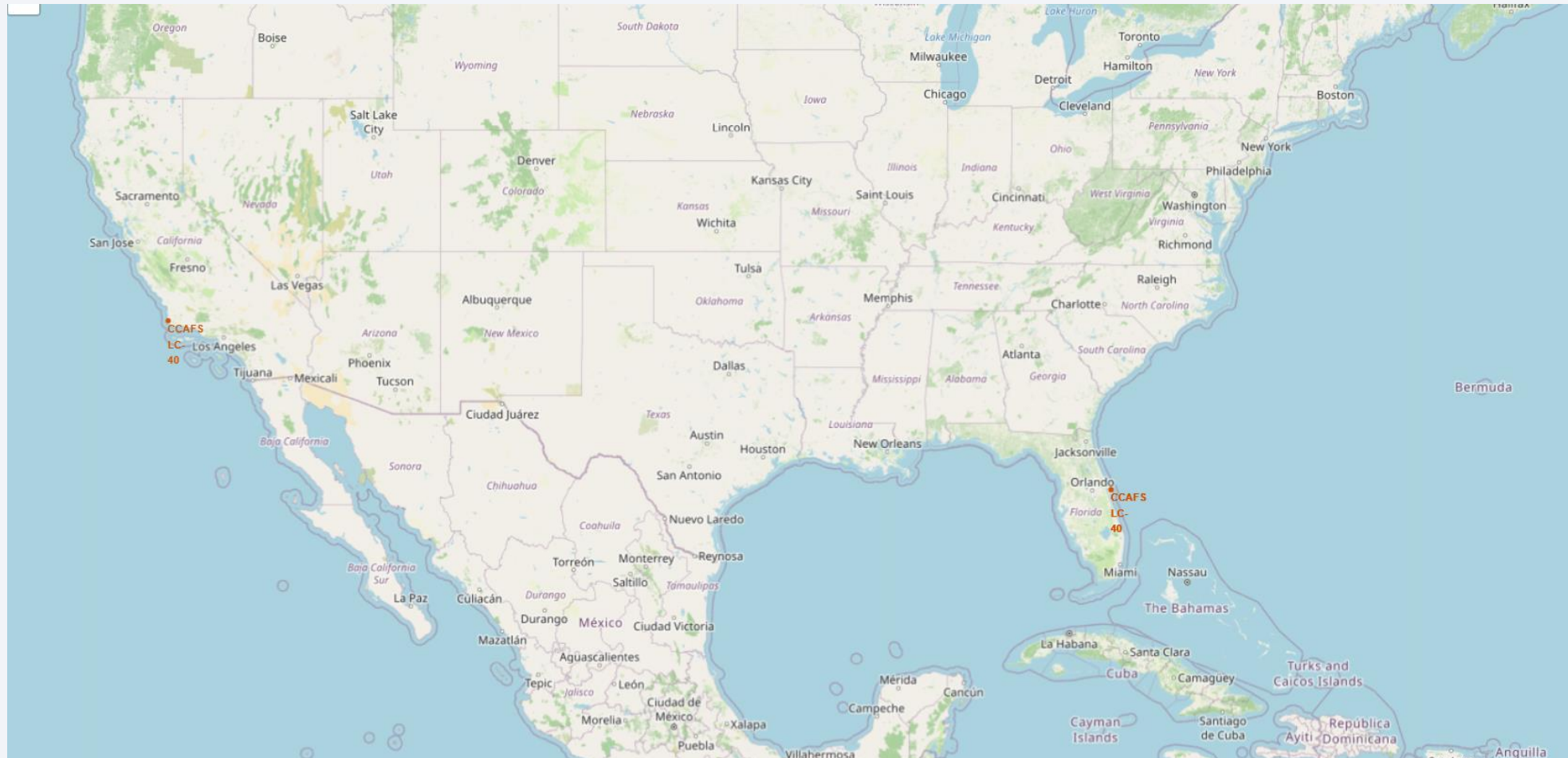
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a solid blue background on the left and a satellite photograph of Earth on the right. The Earth's surface is dark, with numerous bright yellow and orange lights representing cities and urban areas. The horizon of the Earth is visible as a thin, curved line separating the dark surface from the deep blue of space.

Section 3

Launch Sites Proximities Analysis

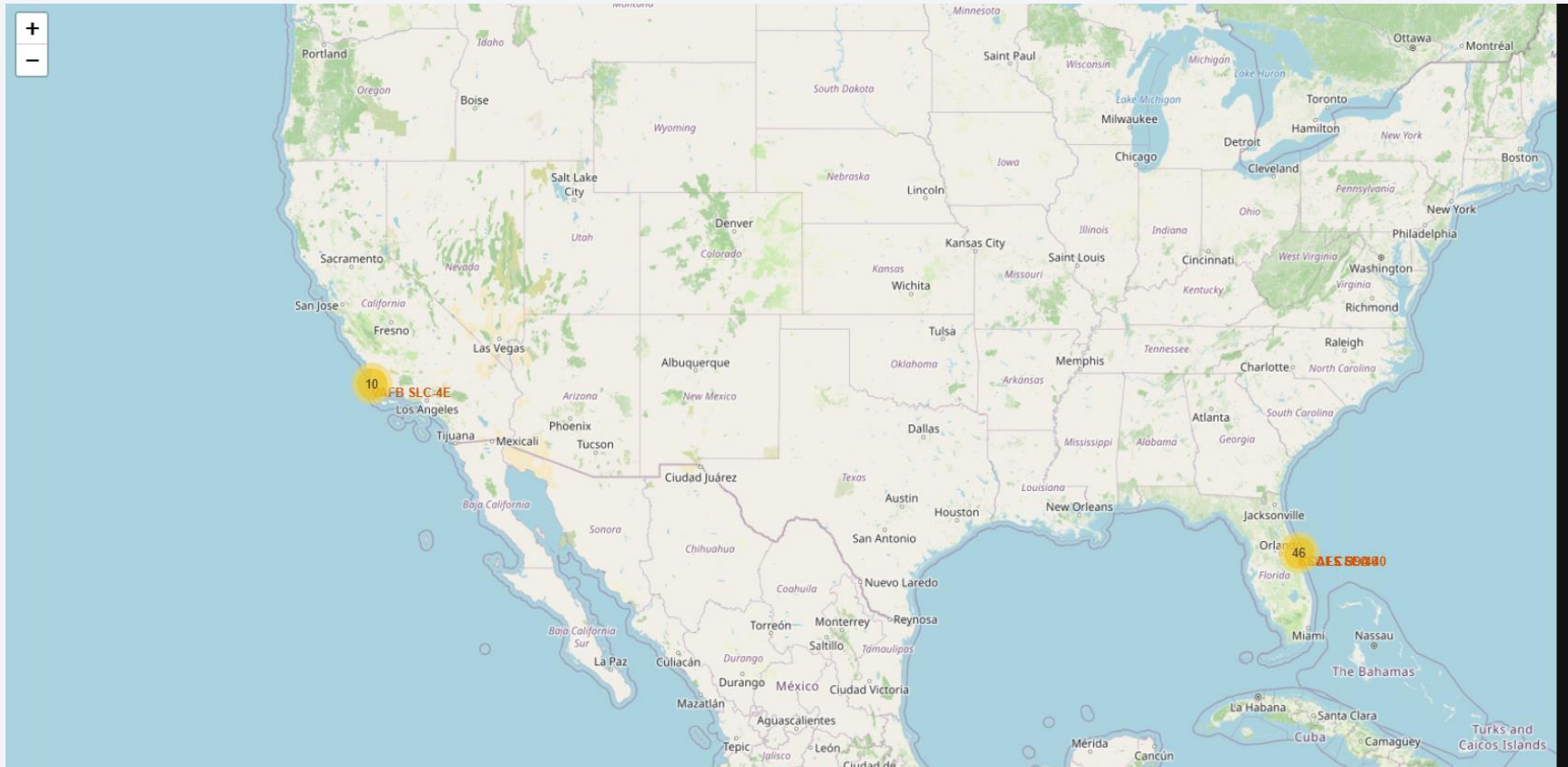
The locations of launch sites

The launch sites are located near the coastlines



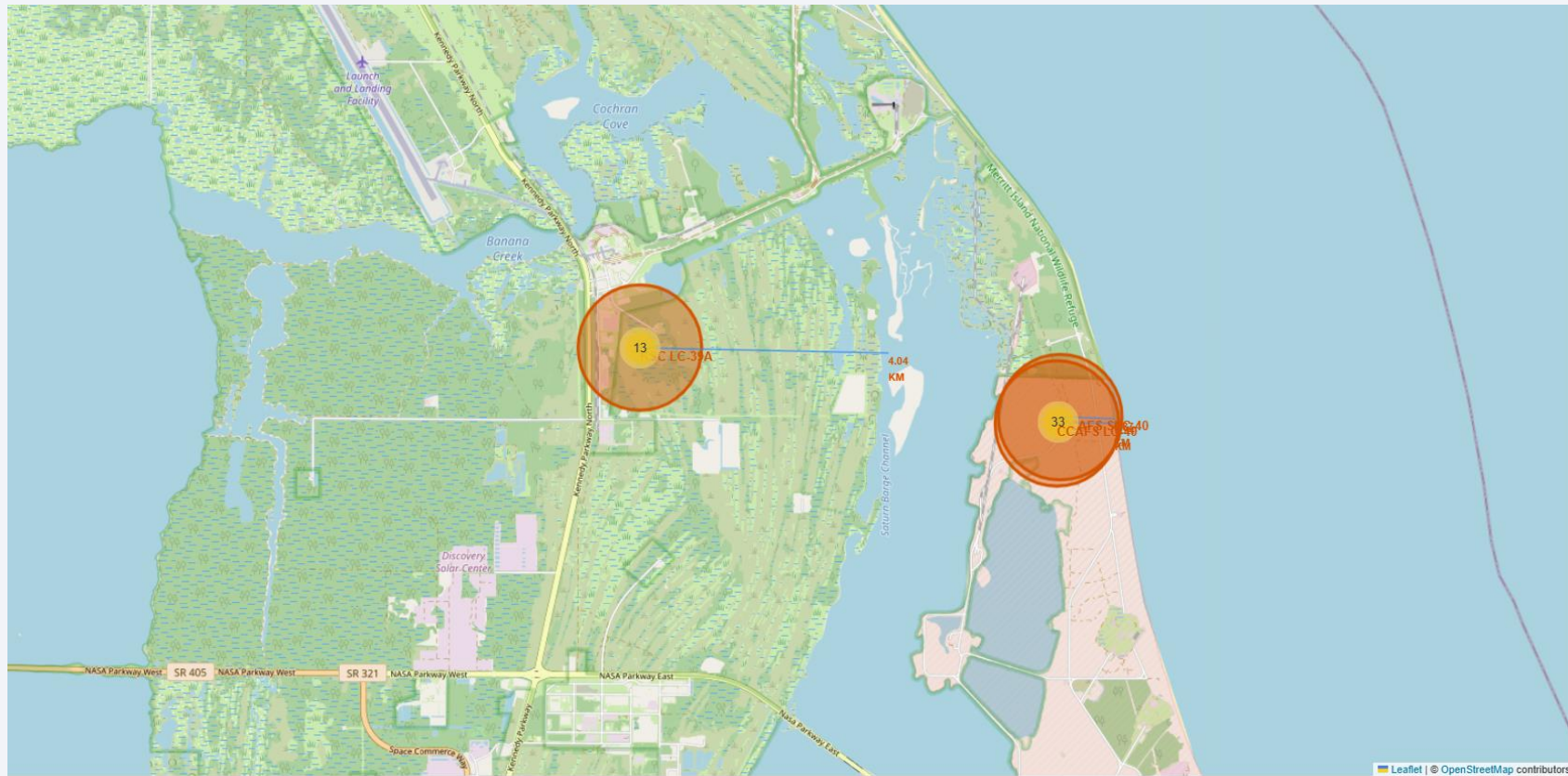
The successful vs. the failure launches on each site

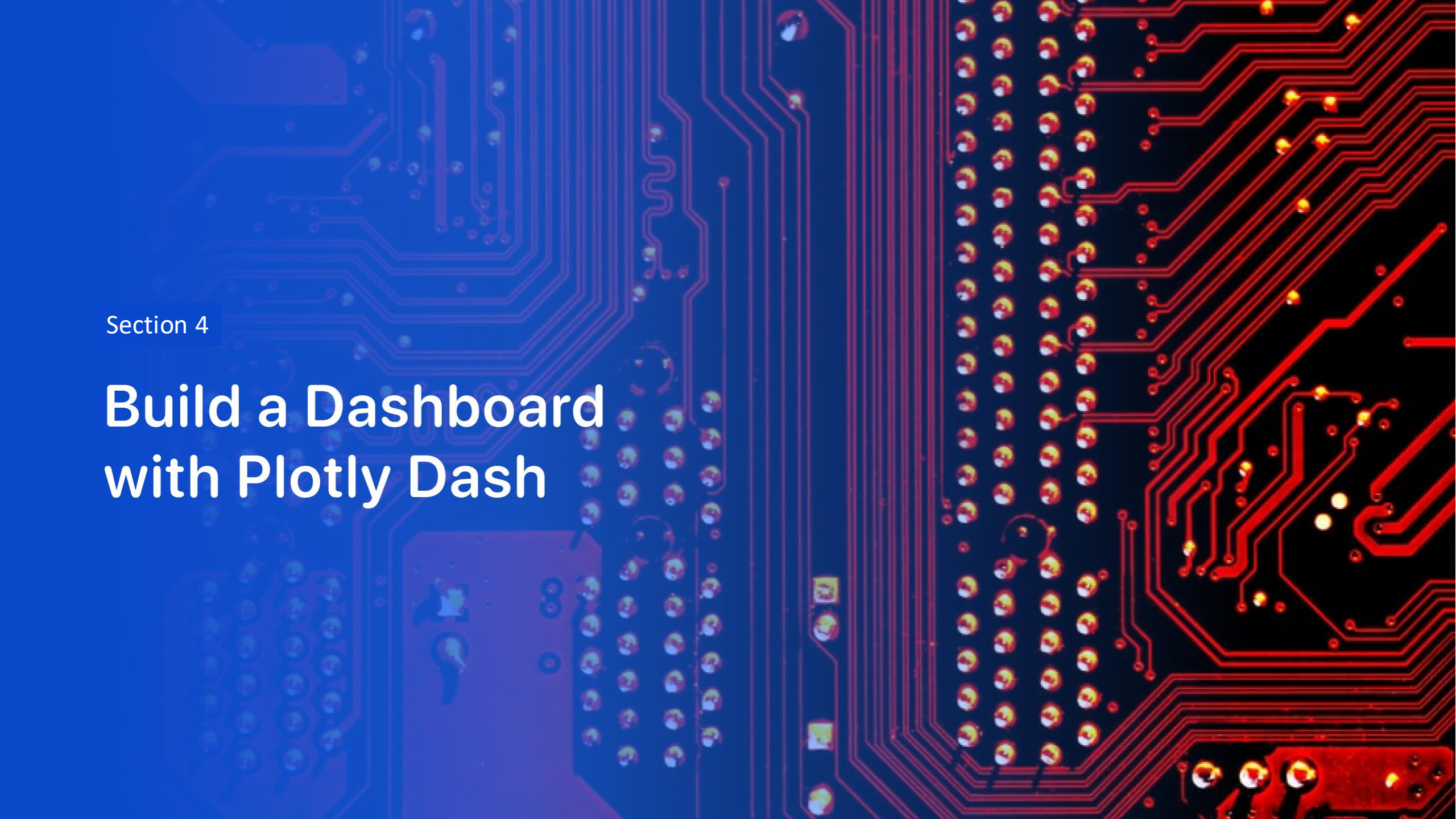
- Site KSC LC-39A has the largest successful rate. CAFS LC-40 has a large failure launches



Distances to the coastlines

- The distances to the coastlines are very small.





Section 4

Build a Dashboard with Plotly Dash

Total Successful Launches by Site

- Site KSC LC-39A has the largest successful rate.

SpaceX Launch Records Dashboard

All Sites



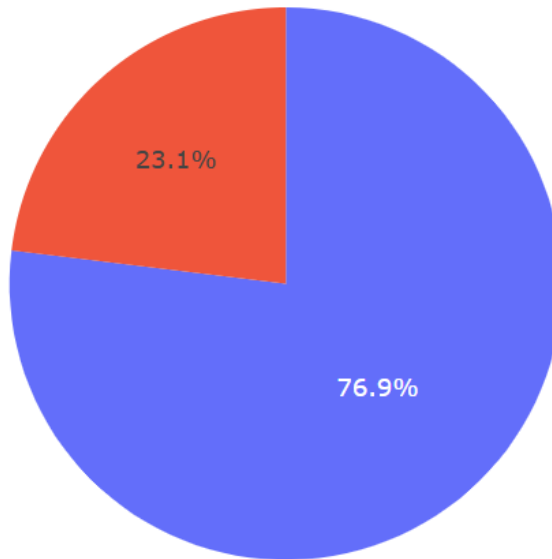
Total Successful Launches by Site



Success vs. Failure for KSC LC-39A

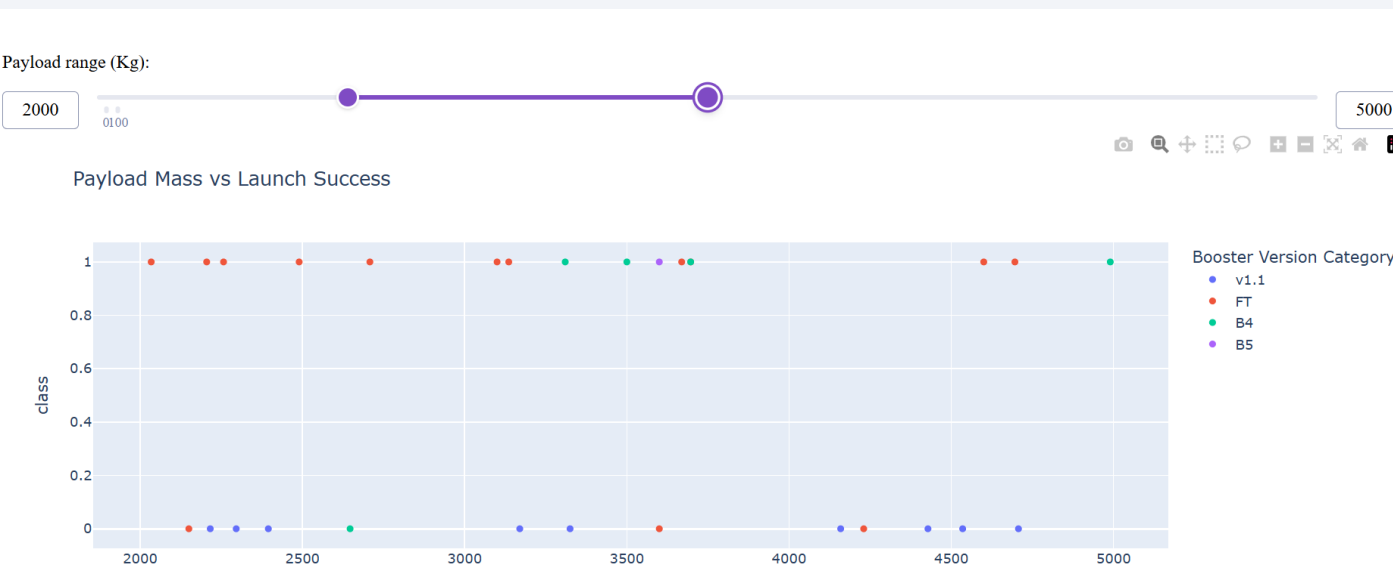
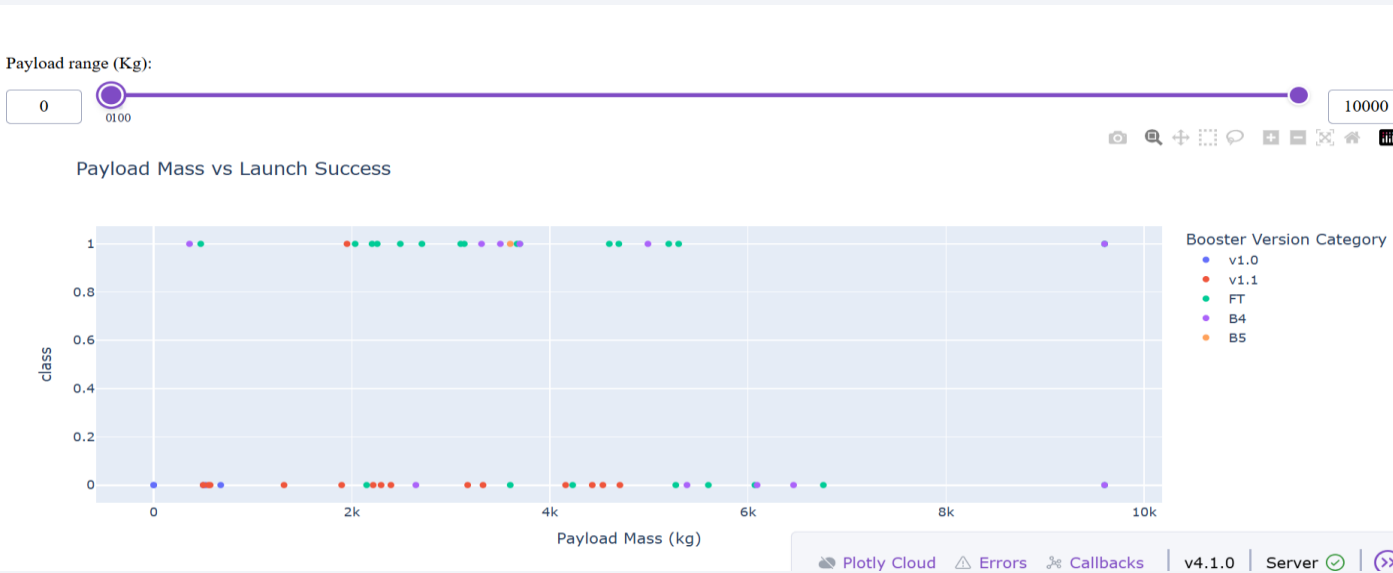
- This site has 76.9% successful rate.

Success vs Failure for KSC LC-39A



Success
Failure

Payload Mass vs Launch Success



The Booster version FT in the payload mass range from 2000kg to 5000kg has the highest successful rate.

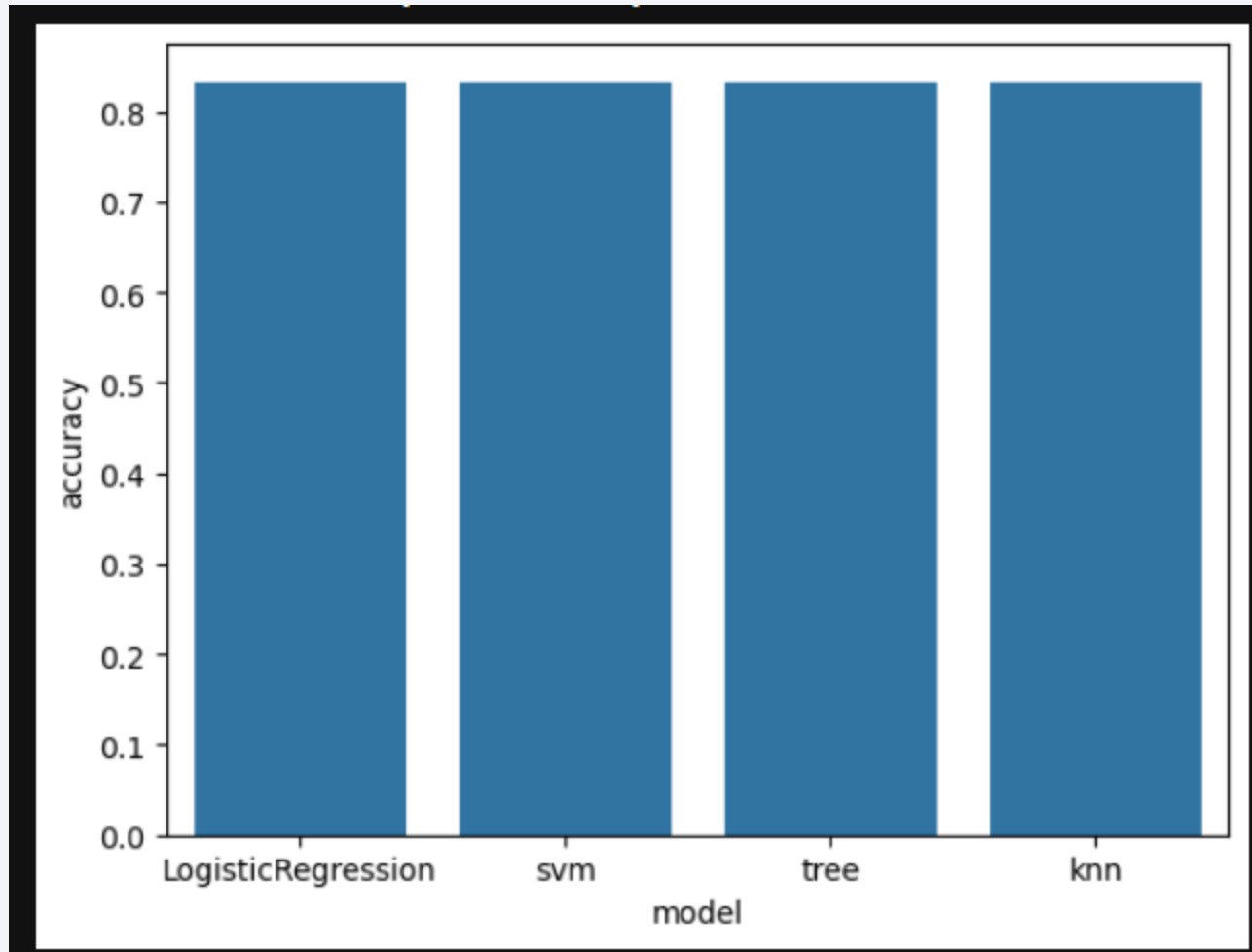


Section 5

Predictive Analysis (Classification)

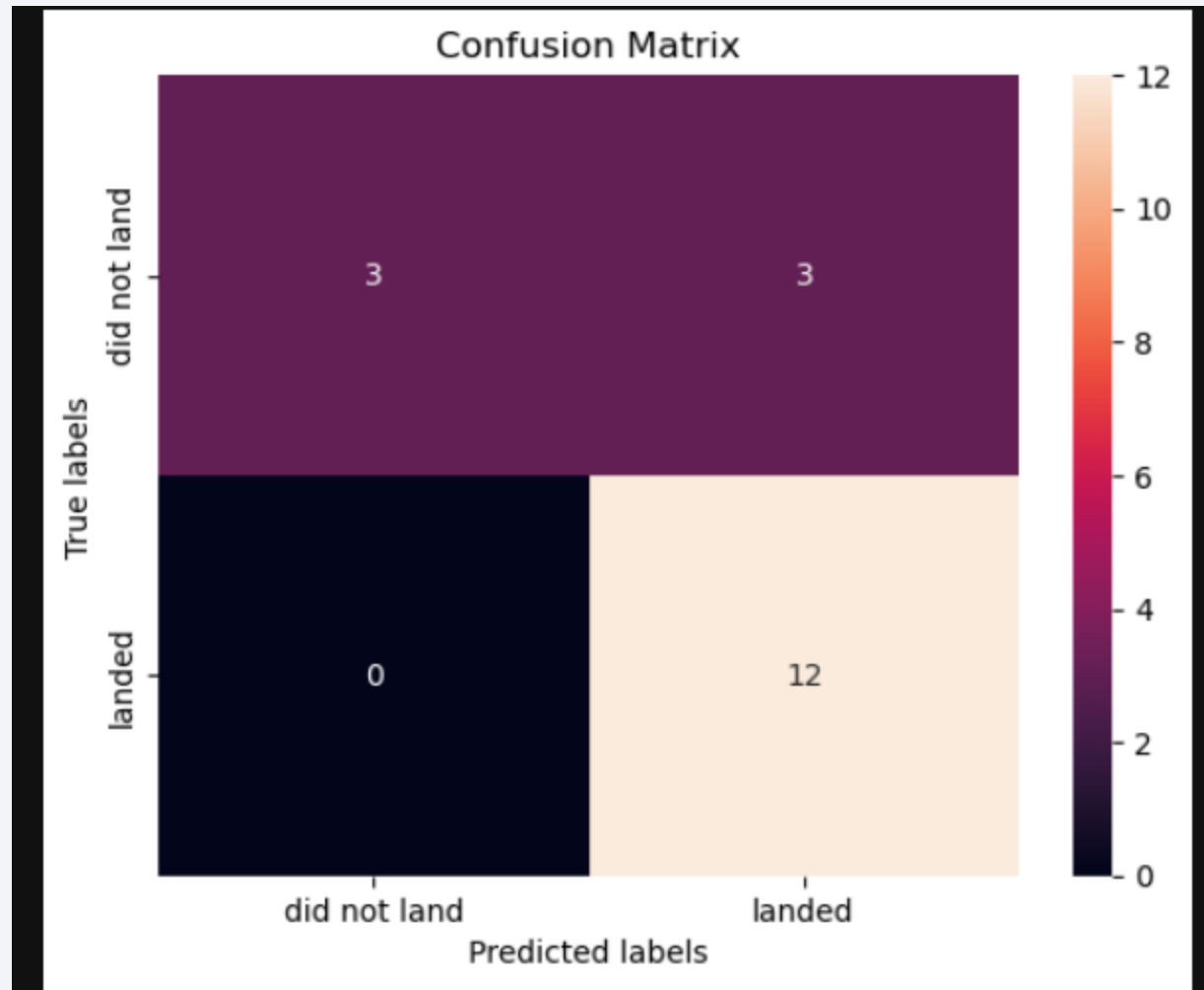
Classification Accuracy

- All models have the same accuracy.



Confusion Matrix

- The model has 12 true positives, 3 false positives.



Conclusions

- We have collected data from the API and Web Scrapping
- We have cleaned the data collected and labeled the successful and failed landings
- We showed that the orbits with the highest successful rates
- We visualized the launch sites' locations on the map
- We created the interactive dashboard to show the successful rate of all launch sites
- We observed that the success rate increases
- We found the model with its predictive accuracy.

Appendix

- The Github link for the project:

https://github.com/nguyenquanghuyvn/Applied_Data_Science_Capstone/tree/main

Thank you!

